



University of
Nottingham

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COMP4139

Machine Learning

- Supervised Nonlinear Models (3)

Artificial Neural Networks

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Recap

- What are non-linear models?
- SVMs in one line?
- SVMs – linear or non-linear?
- SVMs – hard and soft margin?
- Decision Trees in one line?
- Random Forests in one line?

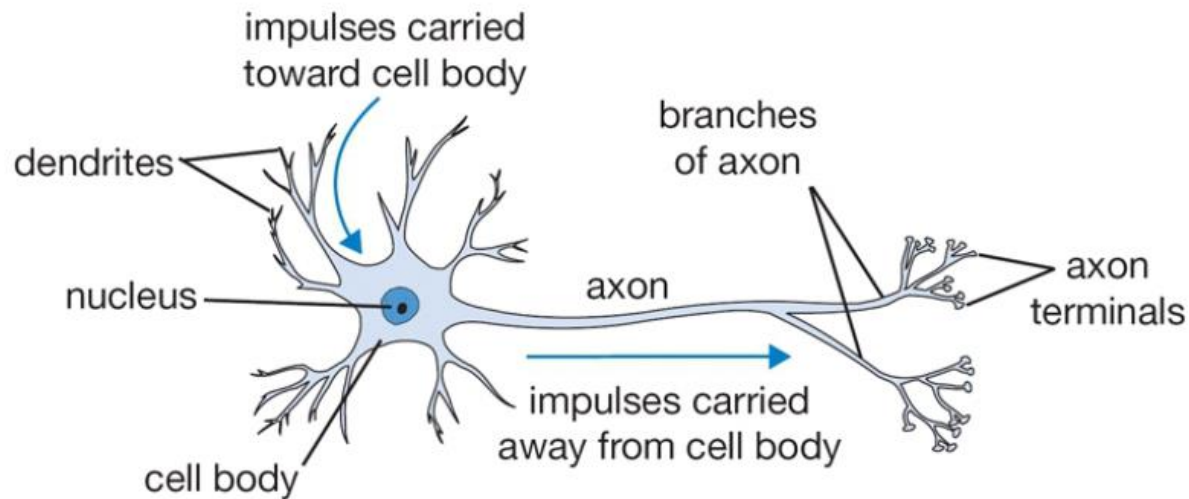


Learning Objectives

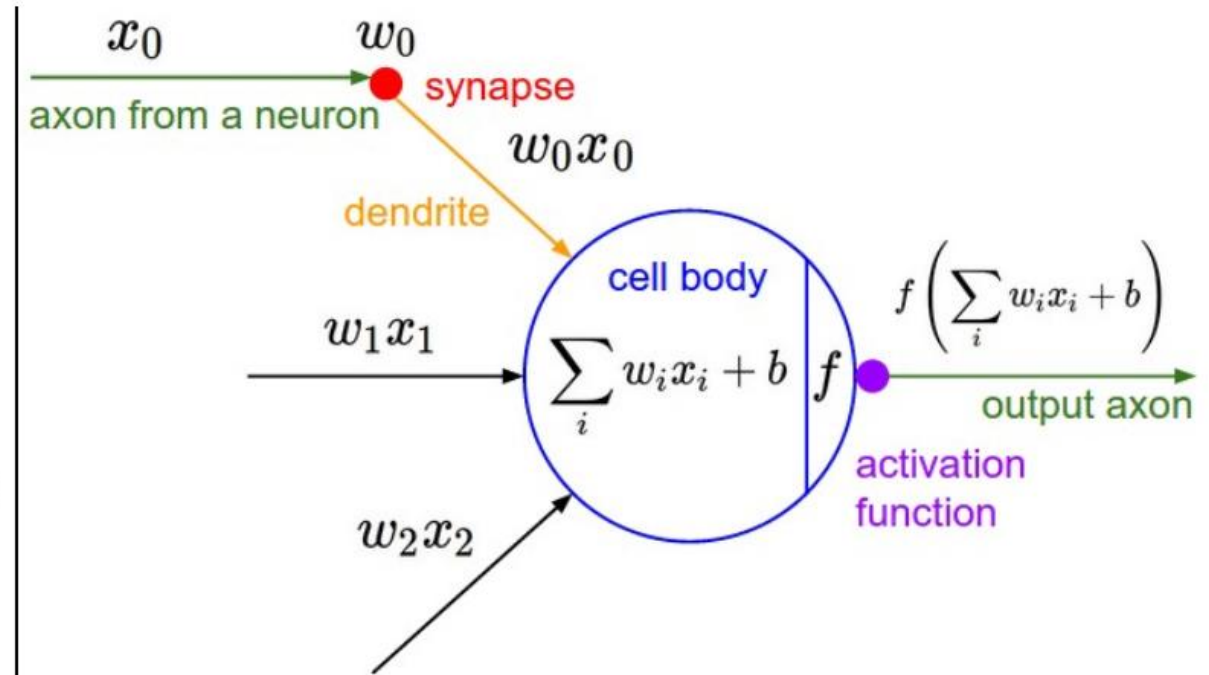
- What is a neural network?
- What are the different layers in a neural network?
- What are activation functions?
- How do we train a neural network?
- Example on backpropagation

Neural Unit

- ~86 billion neurons in human's nervous system
- Connected with 10^{14} - 10^{15} synapse



Biological illustration

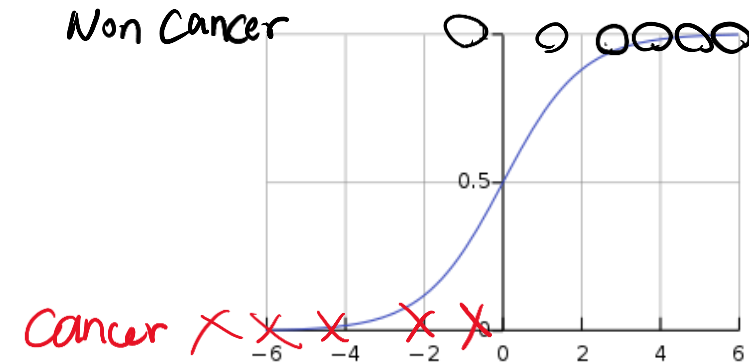
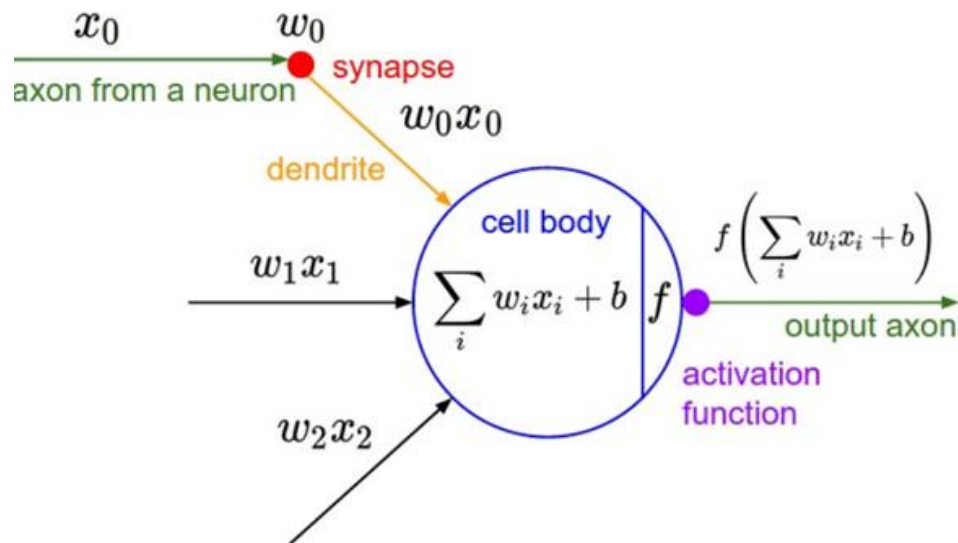


Mathematical model



Neural Network vs Logistic Regression

- Logistic Regression can be implemented using one neuron



Logistic function $P(x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$

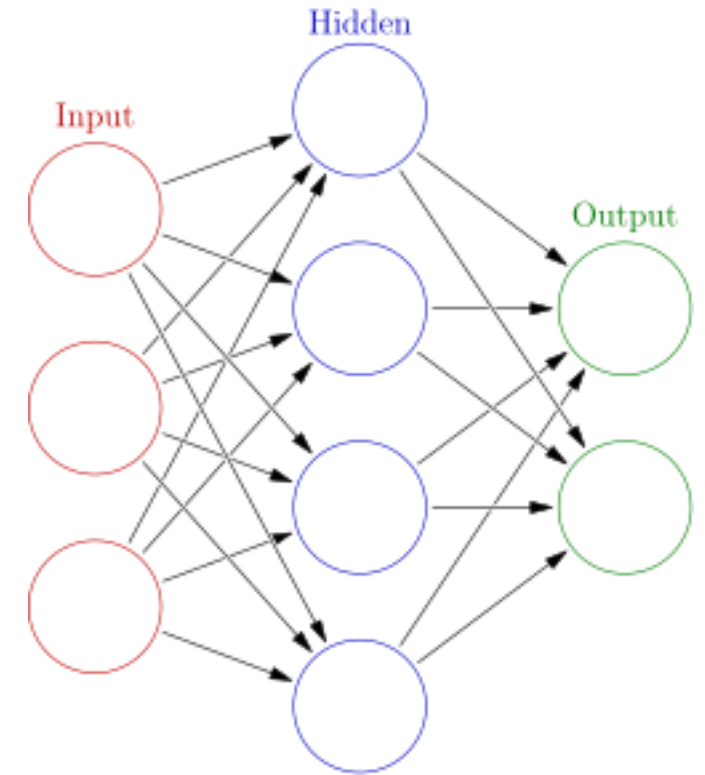
True label

Predicted probability

$$L = \sum_{k=1}^K (y_k \ln(p_k) + (1 - y_k) \ln(1 - p_k))$$

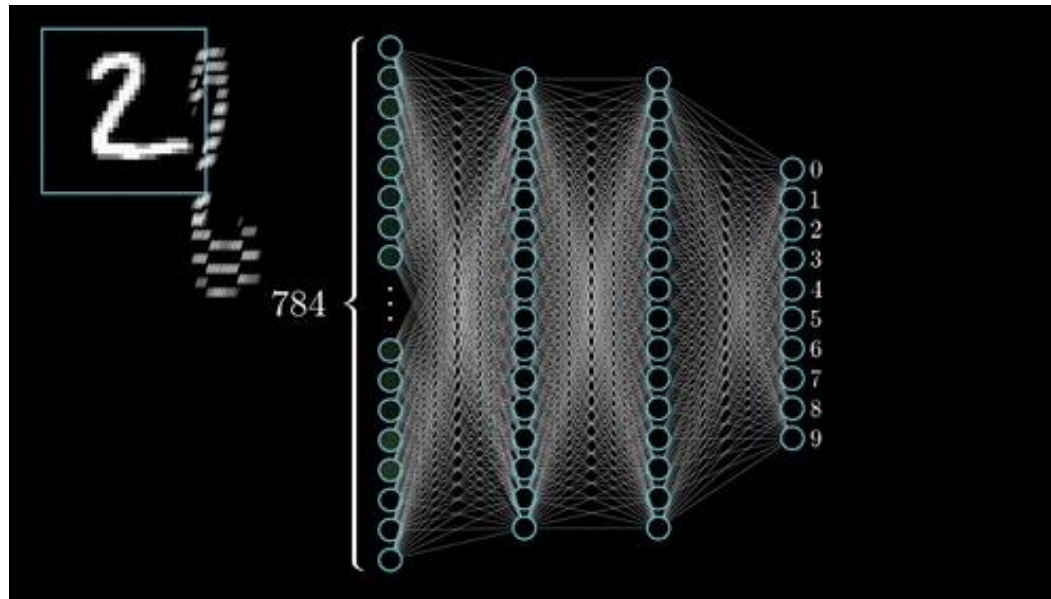
Find the best w_0, w_1 that maximise L or minimise $-L$
Can be minimised using gradient descent.

- **Input layer** connects to the input data (normally features)
- **Hidden layer** performs feature interactions (weighted sum)
- **Output layer** is in charge of the decision making



Multiple Layer Perceptron Neural Network

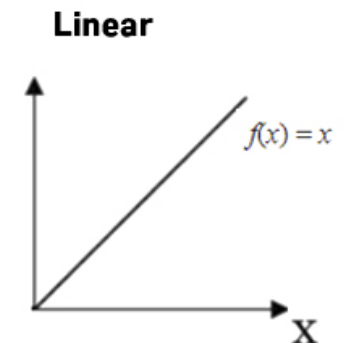
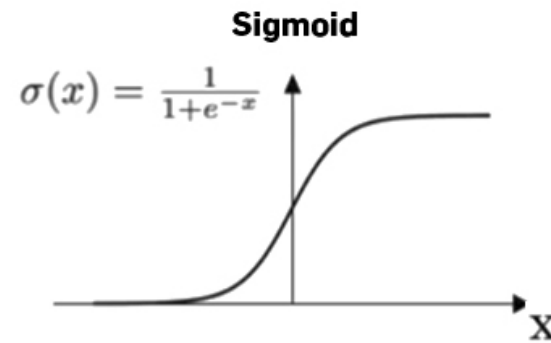
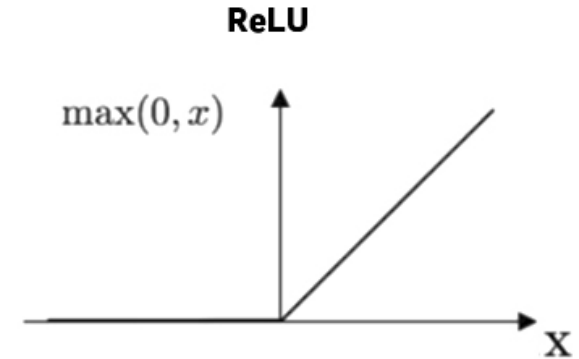
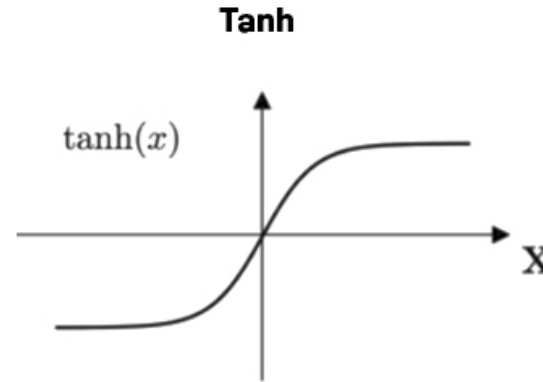
- Stack **lots of neurons** in multiple layers to learn a complex relationship
- A set of **training examples** to train the neural network
- **Weights** that connect neurons are updated to improve the prediction during training
- After training, the network can be used to make predictions of unseen data quickly





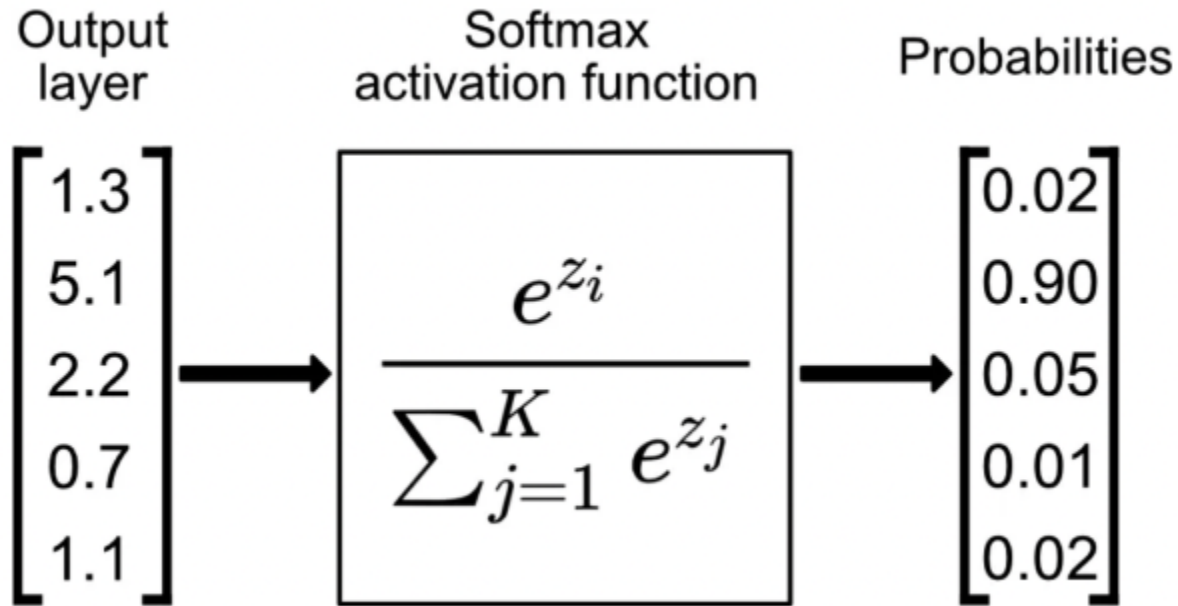
Activation Functions

- Each neurons in the hidden layer and output layer contains an activation function.
- What is an activation function?
- The hidden layer and output layer can have different activation functions. Output layer's activation function is task dependent. (e.g. Sigmoid for binary classification, linear for regression)
- Why do we need activation functions?





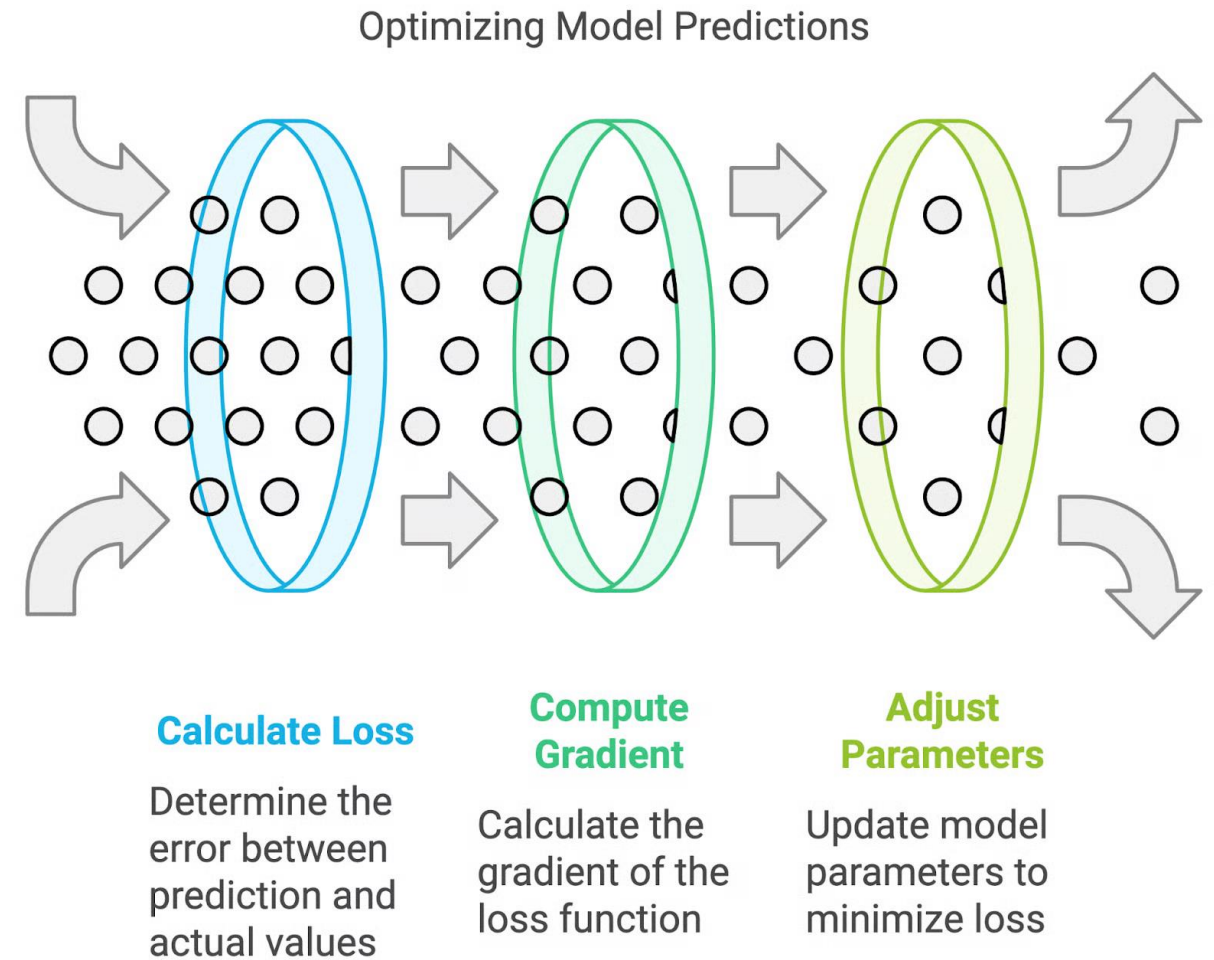
Softmax Activation Function



Raw output values	[3.2, -5.7, 0.6]
Applying sigmoid to raw output values	sigmoid calculation for the first raw output value: $\sigma(3.2) = \frac{e^{3.2}}{1 + e^{3.2}} = 0.96$ result of sigmoid calculation for all three output values: [0.96, 0.0033, 0.65] Sum: $0.96 + 0.0033 + 0.65 = 1.61 \neq 1$
Applying softmax to raw output values	softmax calculation for the first raw output value: $\text{softmax}(3.2) = \frac{e^{3.2}}{e^{3.2} + e^{-5.7} + e^{0.6}} = 0.93$ result of softmax calculation for all three output values: [0.93, 0.00013, 0.069] Sum: $0.93 + 0.069 + 0.00013 = 1$

Loss Functions

- Measure performance – Single numerical value telling us how good/bad a model is doing
- Guide training – Loss is used to guide optimizing weights and biases using gradients
- Define the goal – It defines the problem we are tackling i.e. classification, regression etc.





Loss Functions

Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Penalizes
larger errors more

Use For: Regression

Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Penalizes
all errors equally

Use For: Regression

Huber Loss

$$L_{\delta}(a) = \begin{cases} \frac{1}{2}a^2 & \text{if } |a| \leq \delta \\ \delta(|a| + \frac{1}{2}\delta) & \text{if } |a| > \delta \end{cases}$$
$$= a = y - \tilde{y}$$

Combines MSE and
MAE

Use For: Regression

Binary Cross-Entropy (Log Loss)

$$BCE = -\frac{1}{n} [y \log(\hat{y}_i) \\ + (1 - y_i) \log(1 - \hat{y}_i)]$$

Penalizes confident,
wrong predictions

Use For:
Binary Classification

Categorical Cross-Entropy

$$CCE = -\sum_{i=1}^n y_i \log(\hat{y}_i)$$

Penalizes
misclassified
predictions

Use For: Multi-Class
Classification

Kullback-Leibler Divergence

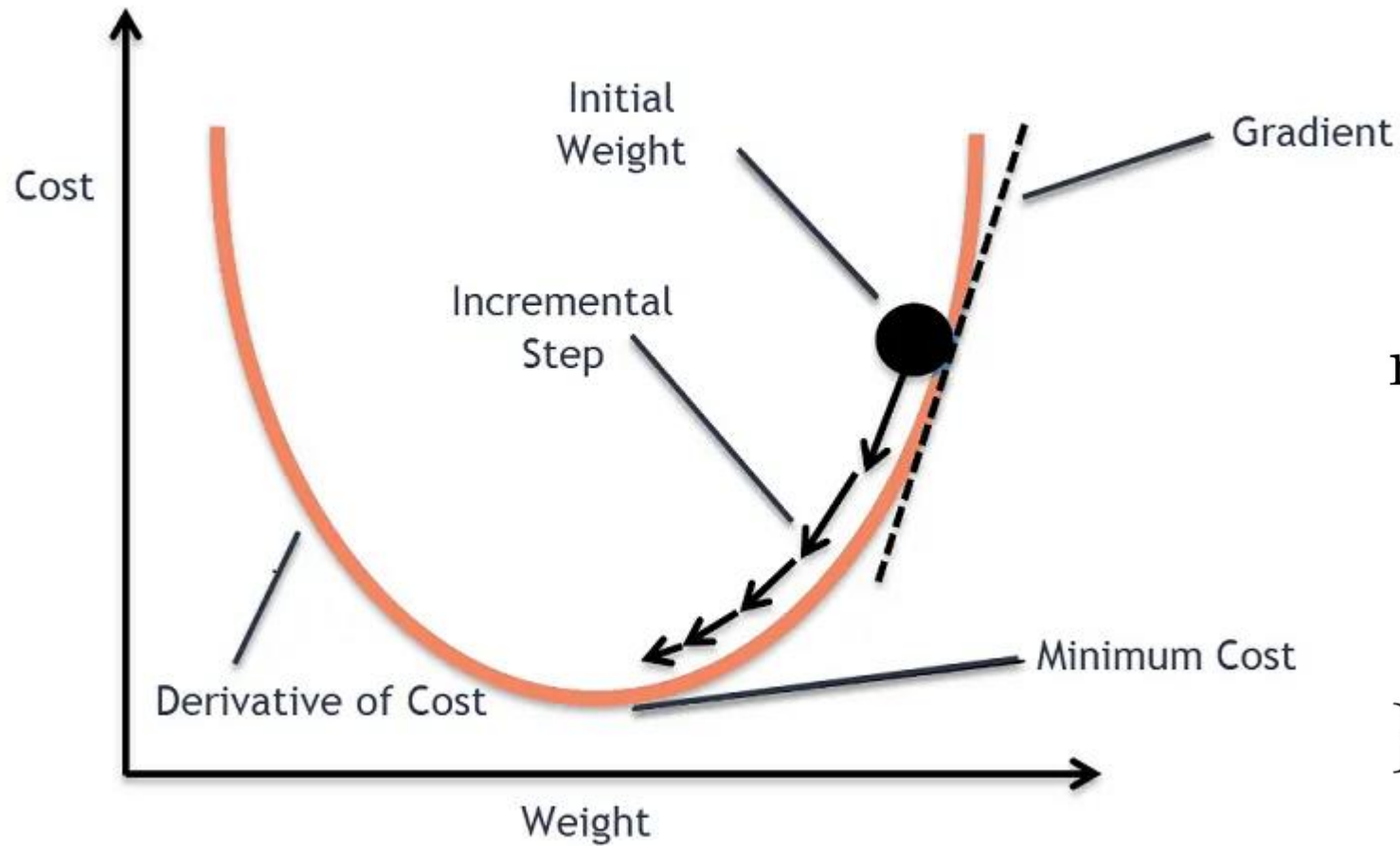
$$D_{KL}(P \parallel Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

Measures
distribution difference

Use For:
Probabilistic Models



Gradient Descent Recap



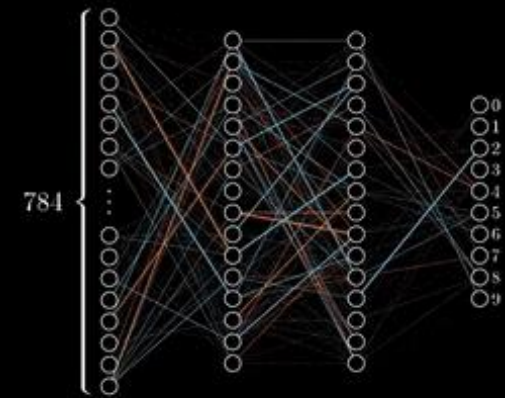
repeat until convergence {
$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(for $j = 1$ and $j = 0$)
}

ANN Training

- Randomly initialise the weights and bias: W and b (*param or hyper-param?*)
- Input data (normally a batch of data one by one) and perform forward passing, then generate a prediction.
- Calculate the error between the predicted output and the true label.
- Backpropagate the error from the output layer back to the input layer by calculating the averaged gradients of the batch for each W and b .
- Update all W and b using the gradient (Gradient Descent).
- Repeat 2-5, until the predicted output is good enough or W and b can't be further updated.

Training in progress...



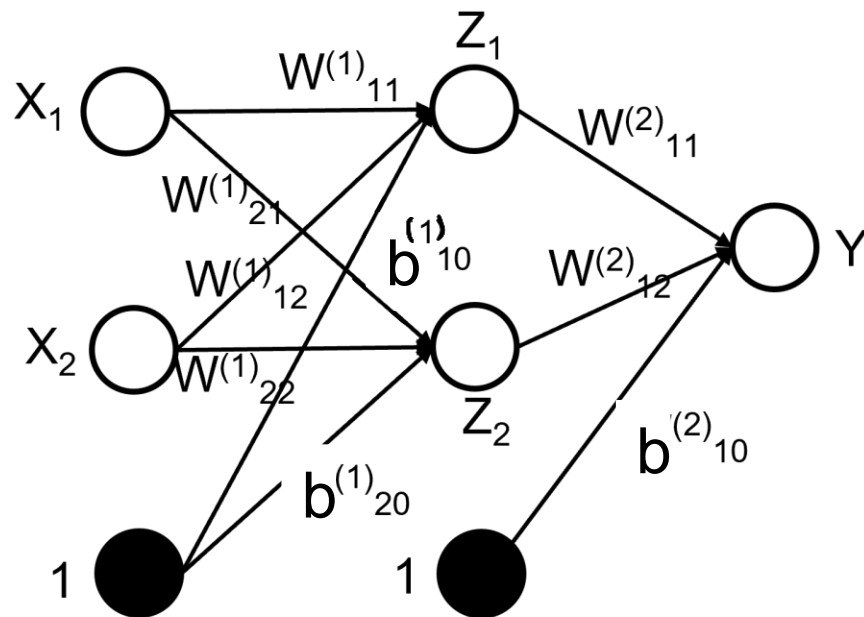
What happens if all W and b are equal?

ANN for Regression (Forward Passing)

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$



$$z_1 = f_z \left(x_1 w_{11}^{(1)} + x_2 w_{12}^{(1)} + b_{10}^{(1)} \right)$$

$$= (x_1 w_{11}^{(1)} + x_2 w_{12}^{(1)} + b_{10}^{(1)})^2 + 3(x_1 w_{11}^{(1)} + x_2 w_{12}^{(1)} + b_{10}^{(1)})$$

$$z_2 = f_z \left(x_1 w_{21}^{(1)} + x_2 w_{22}^{(1)} + b_{20}^{(1)} \right)$$

$$= (x_1 w_{21}^{(1)} + x_2 w_{22}^{(1)} + b_{20}^{(1)})^2 + 3(x_1 w_{21}^{(1)} + x_2 w_{22}^{(1)} + b_{20}^{(1)})$$

$$Y = f_y \left(z_1 w_{11}^{(2)} + z_2 w_{12}^{(2)} + b_{10}^{(2)} \right)$$

$$z_1 w_{11}^{(2)} + z_2 w_{12}^{(2)} + b_{10}^{(2)}$$

Loss function MSE: $L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$ ~ data points

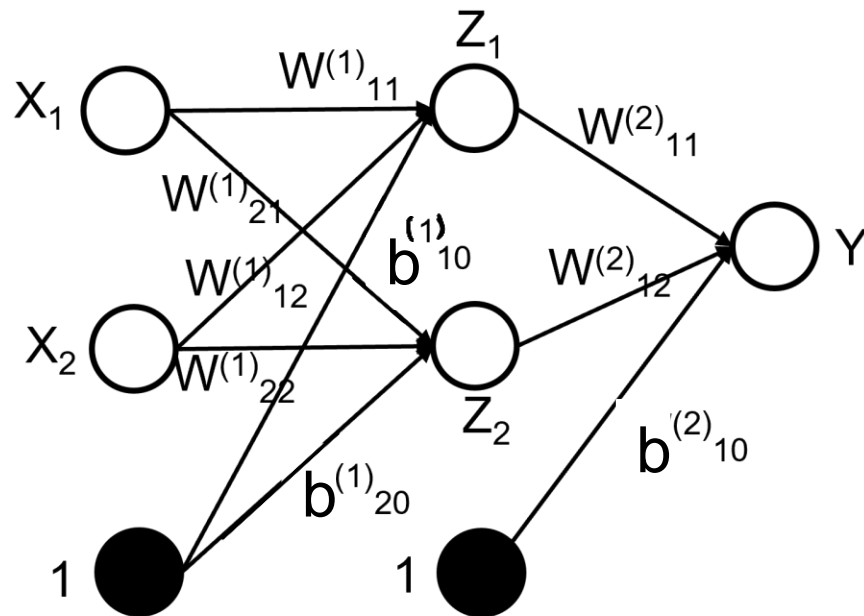
↑ true value ↑ Predicted value

Optimisation (Back Propagation)

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$



Loss function MSE:
$$L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$$

Use gradient descent to update w and b .

How to update $w^{(2)}_{11}$?

$$\frac{\partial L}{\partial w^{(2)}_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial a} \frac{\partial a}{\partial w^{(2)}_{11}}$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n), \quad \frac{\partial \hat{y}}{\partial a} = 1$$

$$\frac{\partial a}{\partial w^{(2)}_{11}} = \frac{\partial (z_1 w^{(2)}_{11} + z_2 w^{(2)}_{12} + b^{(2)}_{10})}{\partial w^{(2)}_{11}} = z_1$$

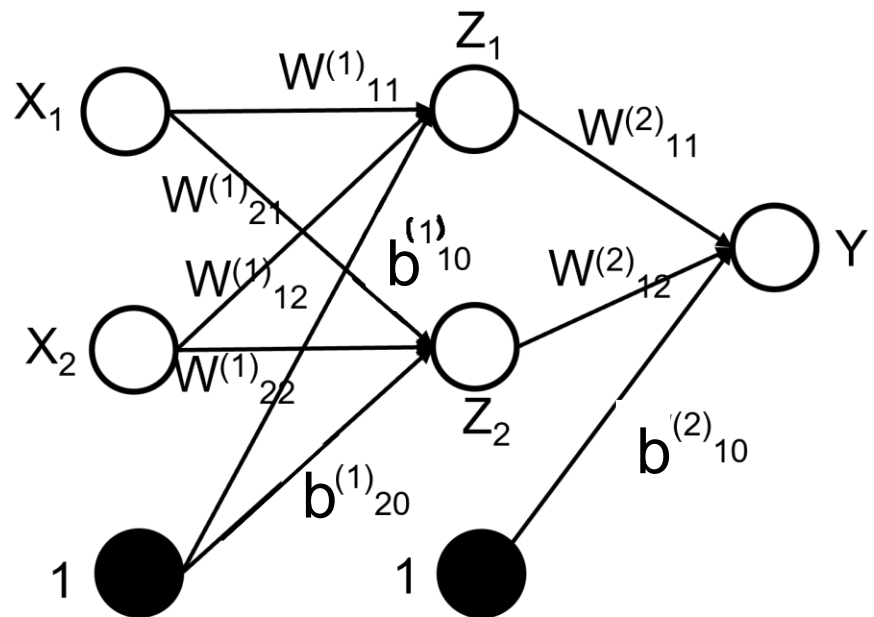
$$\frac{\partial L}{\partial w^{(2)}_{11}} = -\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n) z_1, \quad w^{(2)'}_{11} = w^{(2)}_{11} - \alpha \frac{\partial L}{\partial w^{(2)}_{11}}$$

Example

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$



Parameter

Value

($W^{\{(1)\}}_{11} = 0.10$) ($W^{\{(1)\}}_{12} = 0.20$)

($W^{\{(1)\}}_{21} = 0.30$) ($W^{\{(1)\}}_{22} = 0.40$)

($b^{\{(1)\}}_{10} = 0.05$) ($b^{\{(1)\}}_{20} = -0.05$)

($W^{\{(2)\}}_{11} = 0.25$) ($W^{\{(2)\}}_{12} = -0.30$)

($b^{\{(2)\}}_{10} = 0.10$)

Inputs: $X_1 = 0.05$, $X_2 = 0.10$

Target: $T = 0.30$

Learning rate: $\eta = 0.1$

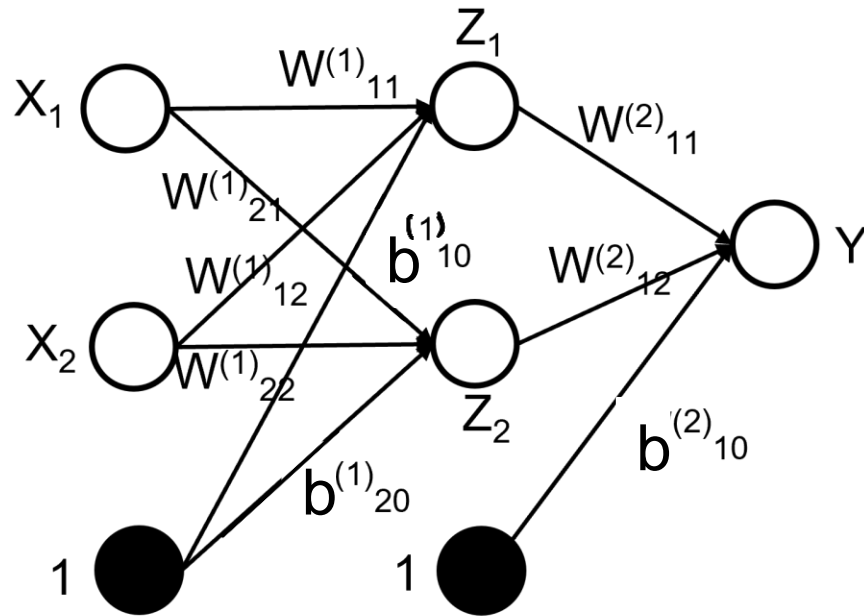
Example

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$.

Hidden Layer



$$a_{Z_1} = X_1 W^{(1)}_{11} + X_2 W^{(1)}_{21} + b^{(1)}_{10}$$

$$a_{Z_1} = 0.05(0.10) + 0.10(0.30) + 0.05 = 0.085$$

$$Z_1 = (a_{Z_1})^2 + 3a_{Z_1} = 0.007225 + 0.255 = 0.2622$$

$$a_{Z_2} = X_1 W^{(1)}_{12} + X_2 W^{(1)}_{22} + b^{(1)}_{20}$$

$$a_{Z_2} = 0.05(0.20) + 0.10(0.40) - 0.05 = 0.03$$

$$Z_2 = (a_{Z_2})^2 + 3a_{Z_2} = 0.0009 + 0.09 = 0.0909$$

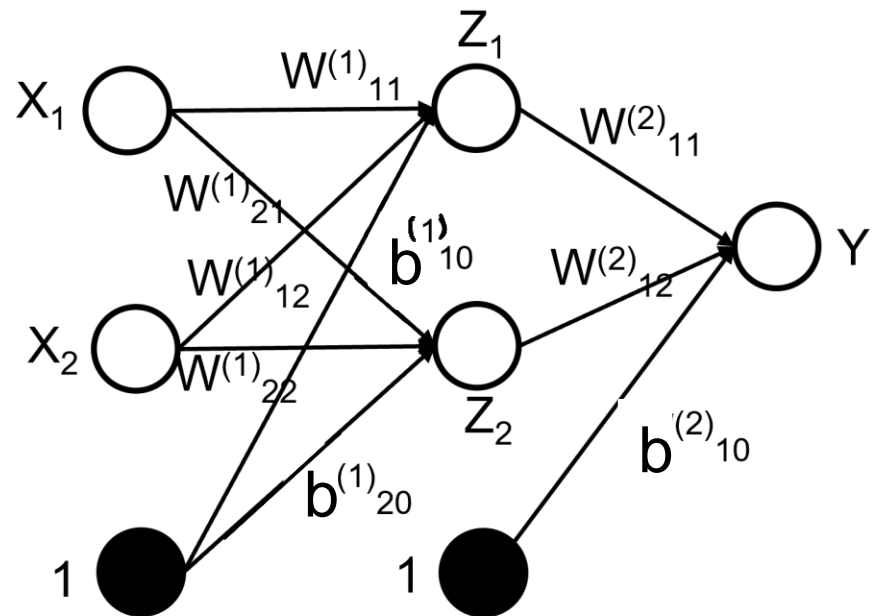
Loss function MSE:
$$L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$$

Example

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$.



Output Layer

$$a_Y = Z_1 W^{(2)}_{11} + Z_2 W^{(2)}_{12} + b^{(2)}_{10}$$

$$a_Y = (0.2622)(0.25) + (0.0909)(-0.30) + 0.10 = 0.06555 - 0.02727 + 0.10 = 0.1383$$

$$Y = f_y(a_Y) = a_Y = 0.1383$$

Loss

$$E = \frac{1}{2}(Y - T)^2 = \frac{1}{2}(0.1383 - 0.30)^2 = 0.01305$$

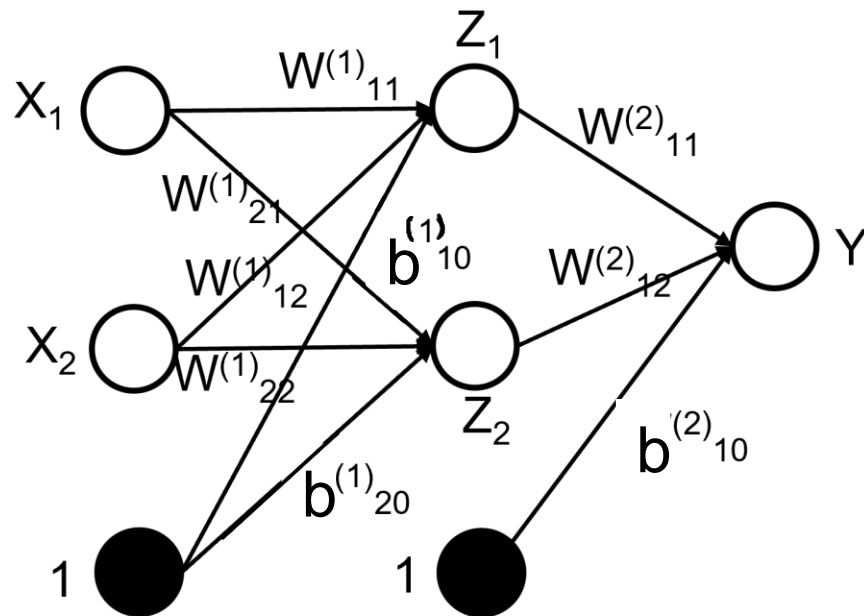
Example – Backward pass

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$

Output Layer



$$\frac{\partial E}{\partial Y} = Y - T = -0.1617$$

$$\frac{\partial E}{\partial W^{(2)}_{11}} = (Y - T)Z_1 = -0.1617(0.2622) = -0.0424$$

$$\frac{\partial E}{\partial W^{(2)}_{12}} = (Y - T)Z_2 = -0.1617(0.0909) = -0.0147$$

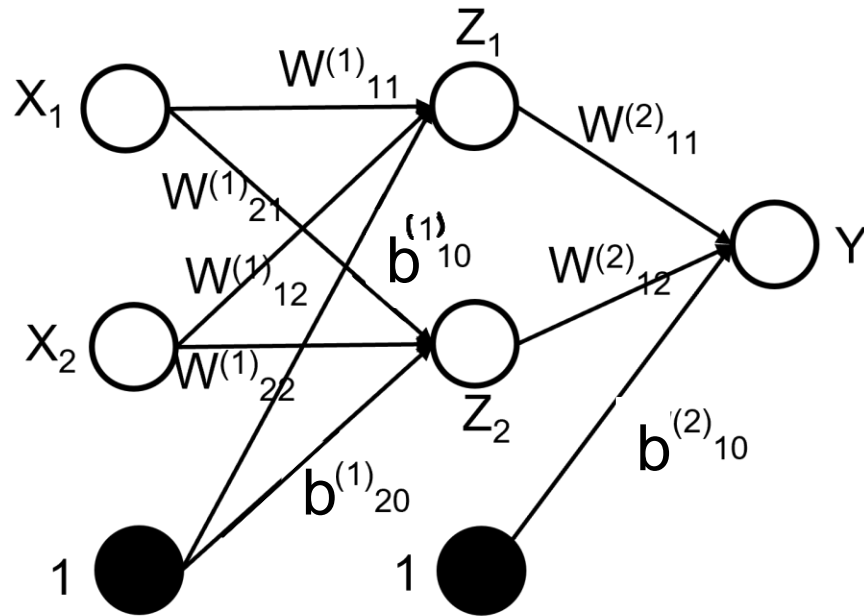
$$\frac{\partial E}{\partial b^{(2)}_{10}} = Y - T = -0.1617$$

Example – Backward pass

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$.



Loss function MSE:
$$L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$$

Hidden Layer

$$\frac{\partial Z_i}{\partial a_{Z_i}} = 2a_{Z_i} + 3$$

For Z_1 :

$$\frac{\partial E}{\partial a_{Z_1}} = (Y - T)W^{(2)}_{11}(2a_{Z_1} + 3)$$

$$\frac{\partial E}{\partial a_{Z_1}} = (-0.1617)(0.25)(3.17) = -0.1288$$

For Z_2 :

$$\frac{\partial E}{\partial a_{Z_2}} = (Y - T)W^{(2)}_{12}(2a_{Z_2} + 3)$$

$$\frac{\partial E}{\partial a_{Z_2}} = (-0.1617)(-0.30)(3.06) = 0.1485$$

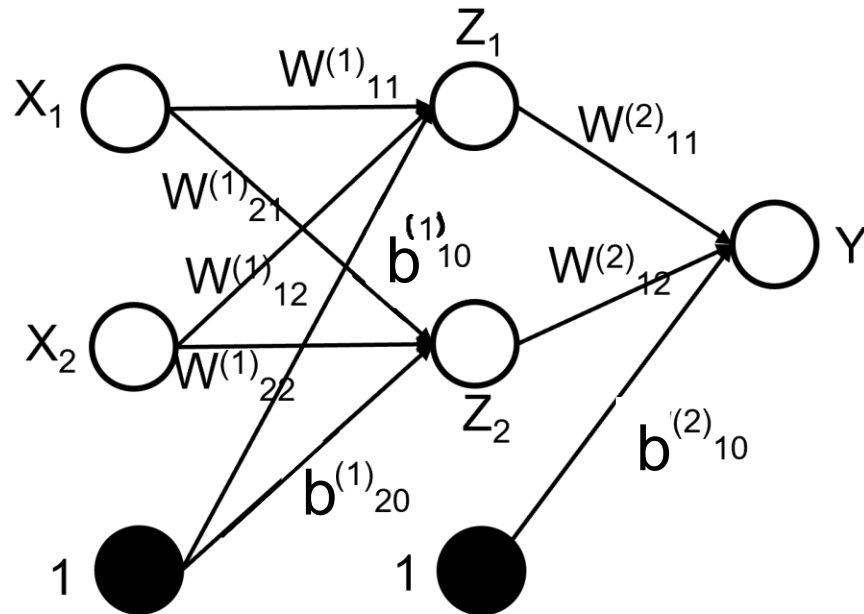
Example – Backward pass

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$

Then:



Loss function MSE:
$$L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$$

$$\frac{\partial E}{\partial W^{(1)}_{11}} = \frac{\partial E}{\partial a_{Z_1}} X_1 = -0.1288(0.05) = -0.00644$$

$$\frac{\partial E}{\partial W^{(1)}_{21}} = -0.1288(0.10) = -0.01288$$

$$\frac{\partial E}{\partial b^{(1)}_{10}} = -0.1288$$

$$\frac{\partial E}{\partial W^{(1)}_{12}} = 0.1485(0.05) = 0.00743$$

$$\frac{\partial E}{\partial W^{(1)}_{22}} = 0.1485(0.10) = 0.01485$$

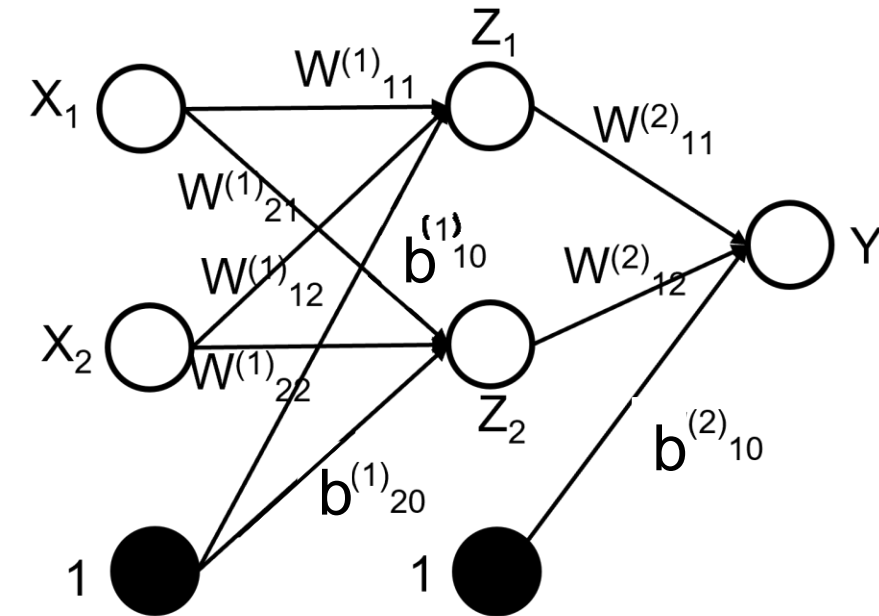
$$\frac{\partial E}{\partial b^{(1)}_{20}} = 0.1485$$



Example – Weight Update

Parameter	Update Rule	New Value
$(W^{(2)}_{11})$	$(0.25 - 0.1(-0.0424))$	0.2542
$(W^{(2)}_{12})$	$(-0.30 - 0.1(-0.0147))$	-0.2985
$(b^{(2)}_{10})$	$(0.10 - 0.1(-0.1617))$	0.1162
$(W^{(1)}_{11})$	$(0.10 - 0.1(-0.00644))$	0.1006
$(W^{(1)}_{21})$	$(0.30 - 0.1(-0.01288))$	0.3013
$(b^{(1)}_{10})$	$(0.05 - 0.1(-0.1288))$	0.0629
$(W^{(1)}_{12})$	$(0.20 - 0.1(0.00743))$	0.1993
$(W^{(1)}_{22})$	$(0.40 - 0.1(0.01485))$	0.3985
$(b^{(1)}_{20})$	$(-0.05 - 0.1(0.1485))$	-0.0649

- How to convert the network on the right to a binary classification network?
- In the multi-class classification ANN, how do we label our data?
- What are the hyperparameters of ANN?
- Compare decision trees and ANN? Pros and Cons.
- Check how to perform partial derivatives of sigmoid function. (chain rule)

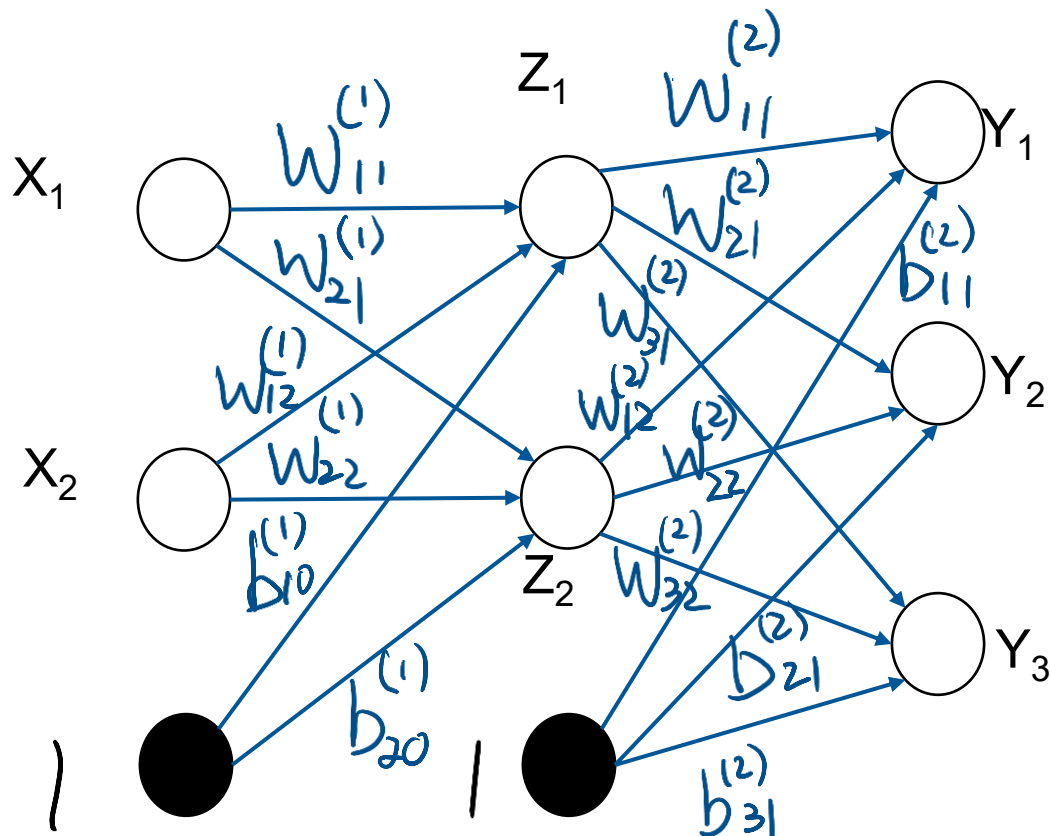


ANN for Multi-class Classification (Forward Passing)

Example:

Activation function at Z: $f_z(a) = a^2 + 3a$

Activation function for Y_1, Y_2, Y_3 are **softmax** functions



$$z_1 = (x_1 w_{11}^{(1)} + x_2 w_{12}^{(1)} + b_{10}^{(1)})^2 + 3(x_1 w_{11}^{(1)} + x_2 w_{12}^{(1)} + b_{10}^{(1)})$$

$$z_2 = (x_1 w_{21}^{(1)} + x_2 w_{22}^{(1)} + b_{20}^{(1)})^2 + 3(x_1 w_{21}^{(1)} + x_2 w_{22}^{(1)} + b_{20}^{(1)})$$

$$\hat{y}_1 = \text{Softmax}(z_1 w_{11}^{(2)} + z_2 w_{12}^{(2)} + b_{11}^{(2)})$$

$$\hat{y}_2 = \text{Softmax}(z_1 w_{21}^{(2)} + z_2 w_{22}^{(2)} + b_{21}^{(2)})$$

$$\hat{y}_3 = \text{Softmax}(z_1 w_{31}^{(2)} + z_2 w_{32}^{(2)} + b_{31}^{(2)})$$

Softmax function:

$$\hat{y}_1 = \frac{e^{c_1}}{\sum_{k=1}^3 e^{c_k}}, \hat{y}_2 = \frac{e^{c_2}}{\sum_{k=1}^3 e^{c_k}}, \hat{y}_3 = \frac{e^{c_3}}{\sum_{k=1}^3 e^{c_k}}, \hat{y}_1 + \hat{y}_2 + \hat{y}_3 = 1$$

Cross entropy loss function:

$$L = - \sum_{n=1}^N \sum_{k=1}^C [y_{kn} \log(\hat{y}_{kn})]$$



What did we learn?

- What is a neural network?
- Neural network vs Logistic Regression
- Different layers in a neural network
- Activation Functions
- Loss Functions
- Backpropagation
- Example of backpropagation



What we will learn in the next lecture

- How do we evaluate ML models?
- What are some commonly used metrics?
- How do we know if the results we have got are statistically significant?